# SQL:
# A COMMERCIAL DATABASE LANGUAGE

## Data Definition,
## Basic Integrity Constraints,
## Schema Changes

# Outline

1. Introduction
2. Data Definition, Basic Constraints, and Schema Changes
3. Basic Queries
4. More complex Queries
5. Aggregate Functions and Grouping
6. Advanced Querying Issues
7. Summary of SQL Queries
8. Data Change statements
9. Views
10. Complex Constraints
11. Database Programming

# 1. Introduction (1)

- The name **SQL** is derived from Structured Query Language. Originally SQL was called SEQUEL (for Structured English QUEry Language).

- SQL is a *comprehensive database language*. It has statements for *data definition*, *query* and *update* (in this sense it is both a *DDL* and a *DML*). In addition, it has facilities for *defining views*, for specifying *security and authorization*, for defining *integrity constraints*, and for specifying *transaction controls*. It also has rules for *embedding SQL statements* into a general-purpose programming language.

# 1. Introduction (1)

- SQL became an *ANSI standard for relational databases* in 1986 (called SQL-86 or SQL1).
  - A revised and more extended version became a standard in 1992 (called SQL-92 or SQL2).
  - A new version of the standard (called SQL:1999) extends SQL with recursive queries, triggers, and some object-oriented features.
  - SQL:2003 introduces XML related features.
  - SQL:2006 introduces more XML and allows applications to integrate SQL with XQuery.
  - SQL:2008 adds INSTEAD OF triggers among other features.
  - SQL:2011 adds temporal data.
  - SQL:2016 adds row pattern matching, JSON etc.

# 1. Introduction (2)

- The SQL language provides a *high-level declarative language* interface: the user specifies what the result is to be and leaves the decisions on how to execute the query to the DBMS.

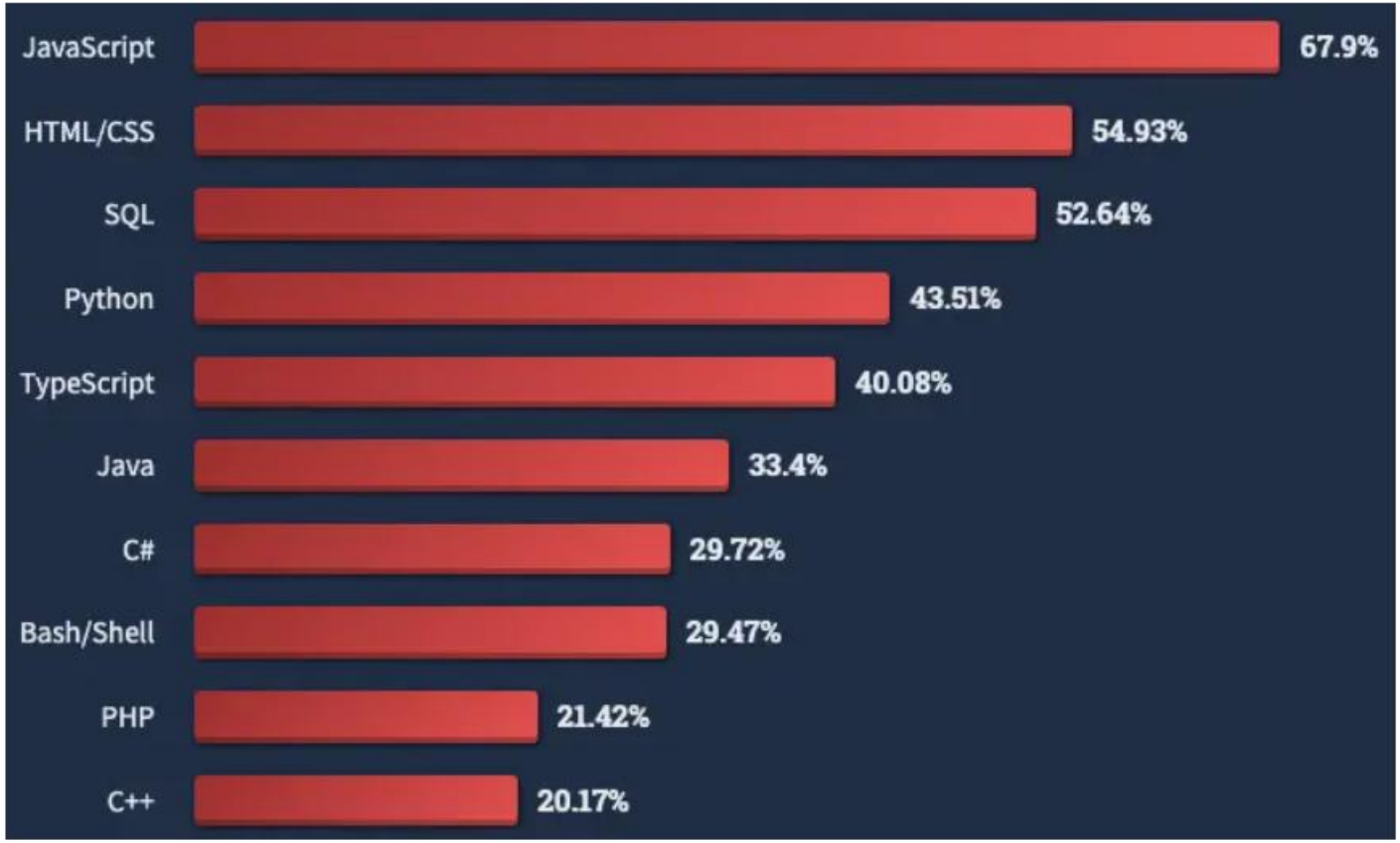
- SQL includes features from *Relational Algebra* but it is based to a greater extend on *Tuple Relational Calculus*. However, Relational Algebra is very important for *query processing and optimization* of SQL queries in relational DBMS.

- In class we will overview the most important features of SQL. Details can be found in SQL documents.

# Popularity of SQL as a programming language in 2022 according to Stack Overflow

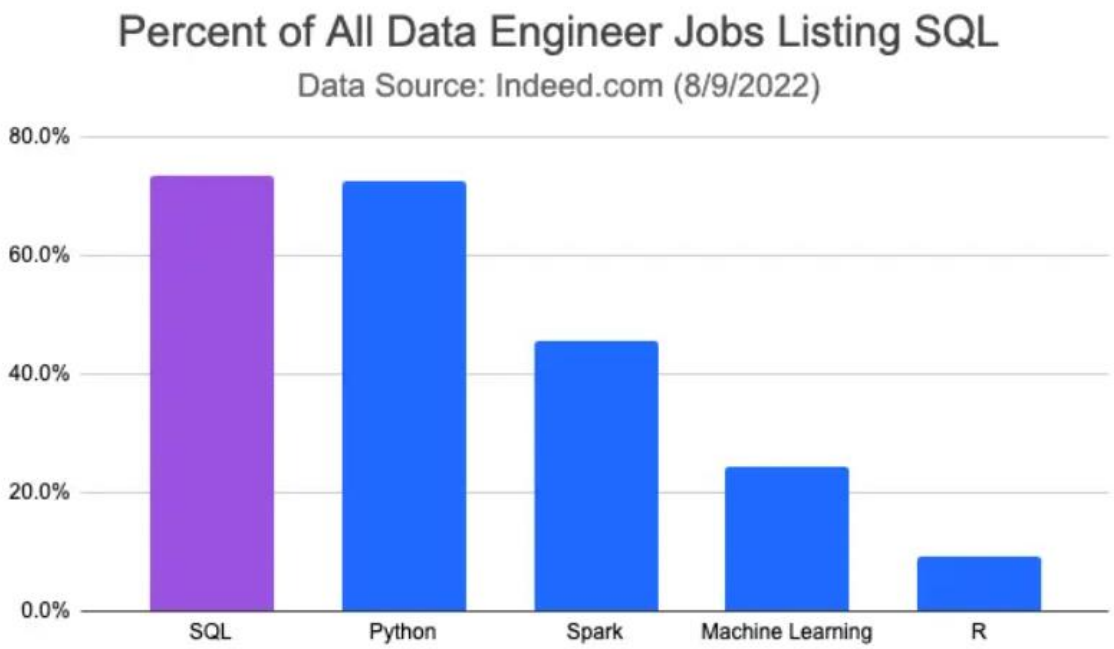
SQL is one of the most-used languages in the entire tech industry!

According to Stack Overflow's 2022 developer survey, SQL eclipses even Python in terms of popularity. In fact, it's the third-most-popular programming language among all professional developers:



# Percent of job postings asking for SQL in 2022 according to the job search engine Indeed.com

**Data Engineer**




Even for data engineering, SQL is the top skill, listed in 73.4% of job postings.

For data engineer roles on Indeed, SQL appears as follows:

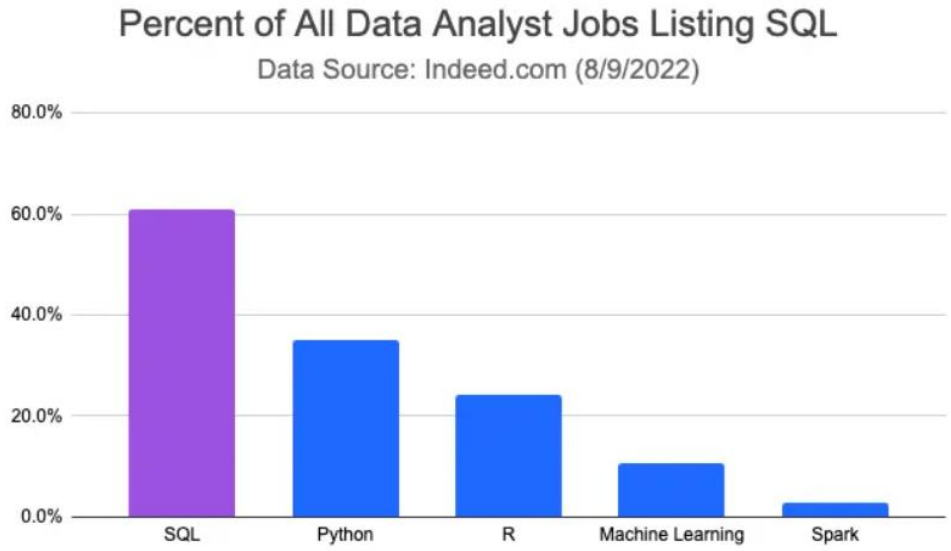
- Slightly more than Python
- 1.6 times more than Spark
- 3 times more than machine learning

# Percent of job postings asking for SQL in 2022 according to the job search engine Indeed.com

**Data Analyst Jobs**

If you're looking for your first job in data, it turns out knowing SQL is even more critical.

Most entry-level jobs in data are data analyst roles, so I analyzed all those on Indeed as well:




For data analyst roles, SQL is again the most in-demand skill, listed in a whopping 61% of job posts.

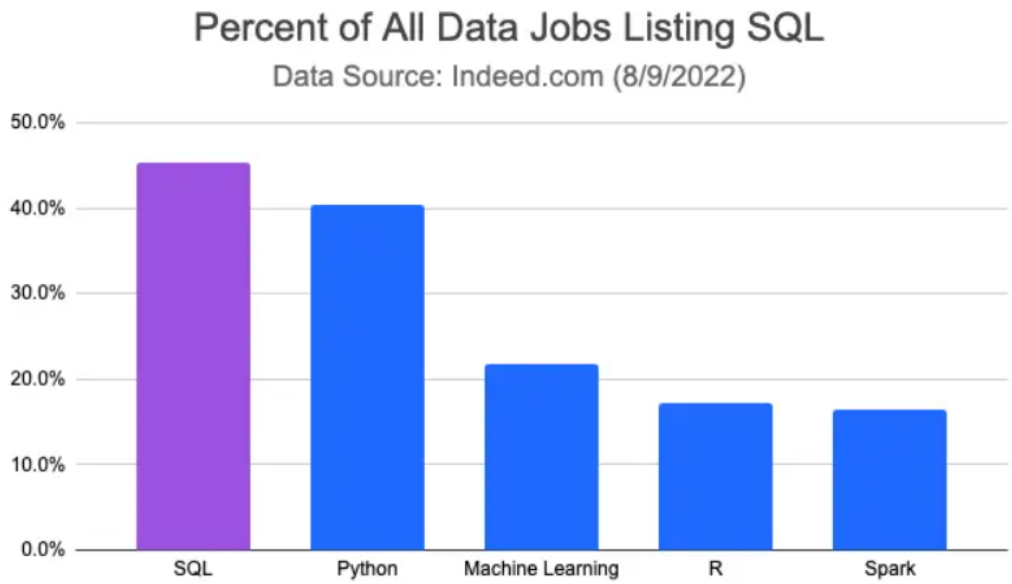
For data analyst roles on Indeed, SQL appears as follows:

- 1.7 times more than Python
- 2.5 times more than R
- 5.8 times more than machine learning
- 22.5 times more than Spark

# Percent of job postings asking for SQL in 2022 according to the job search engine Indeed.com

## All Data Roles

To demonstrate the importance of SQL for data-related jobs, I analyzed more than 72,000 job listings on Indeed, looking at key skills mentioned in job posts with “data” in the title.




As we can see, SQL is the most in-demand skill among all jobs in data, appearing in 45.4% of job postings.

Interestingly, the proportion of data jobs listing SQL has been increasing! I performed this same analysis multiple times over the years, and here’s what I found:

- 2017: 35.7% (SQL #1)
- 2021: 42.7% (SQL #1)
- 2022: 45.4% (SQL #1)

# 1. Introduction (3)

- The terms table, row and column are used in SQL documents for relation, tuple and attribute respectively.

- *SQL is case insensitive*. It treats upper- and lower-case letters as the same letter. Only inside quotes does SQL make a distinction between upper- and lower-case letters.

# 2. Data Definition, Constraints and Schema Changes in SQL

- In this section we discuss DDL commands in SQL.

# 2.1.Schema and Catalog concepts (1)

- The concept of a **schema** is used in SQL to group together constructs that belong to the same database application

- A schema includes **schema elements** such as *tables, constraints, views, domains, authorization grants* and other constructs.

- An SQL schema is identified by a **schema name** and includes an **authorization identifier** to indicate the user who owns the schema. In addition, it can include **descriptors** for every element in the schema.

- The following statement creates a **schema** called COMPANY, owned by a **user** with authorization identifier JSMITH.

**CREATE SCHEMA** COMPANY **AUTHORIZATION** JSMITH;

# 2.1.Schema and Catalog concepts (2)

- The concept of a **catalog** is used in SQL to denote a named collection of schemas.

- A catalog always contains a special schema called INFORMATION_SCHEMA. This schema provides information on all the schemas in the catalog and all the element descriptors in these schemas.

- Some elements (e.g. domain definitions) *can be shared by schemas in the same catalog*.

  Integrity constraints can be defined between relations *only if they exist in schemas of the same catalog*.

# 2.2. Data Types (1)

- The data types available for attribute values include numeric, character-string, bit-string, date and time.
- **Numeric**:
  INTEGER or INT: signed integer
  SMALLINT: signed integer

  FLOAT(p): floating point number
  REAL, DOUBLE PRECISION: alternative spellings for FLOAT(p)

  DECIMAL(i,j) or DEC(i,j) or NUMERIC(i,j): decimal number
- **Character string**:
  CHARACTER(n) or CHAR(n): fixed-length character string
  VARCHAR(n): varying length character string
- **Bit string**:
  BIT(n): fixed-length bit string
  BIT VARYING(n): varying length bit string

# 2.2. Data Types (2)

- **Date and time:**

  DATE: Made up of year-month-day in the format yyyy-mm-dd

  TIME: Made up of hour:minute:second in the format hh:mm:ss

  TIME(i): Made up of hour:minute:second plus i additional digits specifying fractions of a second.
  Format is hh:mm:ss:ii…i

  TIMESTAMP: Has both DATE and TIME components

  INTERVAL: Specifies a relative value rather than an absolute value
  Can be DAY/TIME intervals or YEAR/MONTH intervals
  Can be positive or negative

# 2.3. Domains (3)

- In SQL it is possible to declare a **domain**.

  This facilitates changing the data type for a domain used by numerous attributes, and improves readability.

- Statement:

  **CREATE DOMAIN** SSN_TYPE **AS** CHAR(9);

- A create domain statement can have an optional *default specification* and an optional *list of constraints*.

# 2.4. CREATE TABLE Statement (1)

- The **CREATE TABLE** command is used to specify a new relation, its attributes, their data type and constraints.
- The key, entity integrity and referential integrity constraints can be specified within a CREATE TABLE statement or later using the ALTER TABLE command.

```
CREATE TABLE   EMPLOYEE
  (FNAME        VARCHAR(15)        NOT NULL,
   MINIT        CHAR,
   LNAME        VARCHAR(15)        NOT NULL,
   SSN          CHAR(9)            NOT NULL,
   BDATE        DATE,
   ADDRESS      VARCHAR(30),
   SEX          CHAR,
   SALARY       DECIMAL(10,2),
   SUPERSSN     CHAR(9),
   DNO          INTEGER            NOT NULL,
   PRIMARY KEY (SSN),
   FOREIGN KEY (DNO) REFERENCES DEPARTMENT(DNUMBER),
   FOREIGN KEY (SUPERSSN) REFERENCES   EMPLOYEE(SSN) );
```

# 2.4. CREATE TABLE Statement (2)

- An example with attribute default values and secondary key specification.

```
CREATE TABLE   DEPARTMENT
  (DNAME              VARCHAR(10)   NOT NULL,
  DNUMBER             INT                NOT NULL,
  MGRSSN              CHAR(9)  NOT NULL  DEFAULT '888665555',
  MGRSTARTDATE  DATE,
  PRIMARY KEY (DNUMBER),
  UNIQUE (DNAME),
  FOREIGN KEY (MGRSSN) REFERENCES EMPLOYEE );
```

# 2.4. CREATE TABLE Statement (3)

- A constraint can be given a name following the keyword **CONSTRAINT**. Giving names to constraints is optional but is a good practice to follow.
  The names of all constraints within a particular schema must be unique.
  A constraint name can be used to drop a constraint (and replace it later by another constraint).
- SQL has a default policy that *any modifications violating the referential integrity constraints is rejected by the system*.

  However, in SQL the schema designer can specify the action to be taken upon *deletion of a referenced tuple* or *upon modification of a referenced primary key value*, by attaching a referential triggered action clause to a foreign key constraint.

  The options include SET NULL, CASCADE, SET DEFAULT.

  They can be qualified on either ON DELETE or ON UPDATE.

# 2.4. CREATE TABLE Statement (4)

- Example.

```
CREATE TABLE EMPLOYEE

  (FNAME        VARCHAR(15)        NOT NULL,
   …

   DNO          INTEGER    NOT NULL    DEFAULT  4,

  CONSTRAINT EMPPK
  PRIMARY KEY (SSN),

  CONSTRAINT EMPSUPERFK
  FOREIGN KEY (SUPERSSN) REFERENCES EMPLOYEE
  ON DELETE SET NULL ON UPDATE CASCADE

  CONSTRAINT EMPDEPTFK
  FOREIGN KEY (DNO) REFERENCES DEPARTMENT
  ON DELETE SET DEFAULT ON UPDATE CASCADE );
```

# 2.4. CREATE TABLE Statement (5)

- Example.

```
CREATE TABLE   DEPARTMENT
  (DNAME            VARCHAR(10) NOT NULL,
  DNUMBER            INT                NOT NULL,
  MGRSSN            CHAR(9)          NOT NULL   DEFAULT '888665555',
  MGRSTARTDATE  DATE,

  CONSTRAINT DEPTPK
  PRIMARY KEY (DNUMBER),

  CONSTRAINT DEPTSK
  UNIQUE (DNAME),

  CONSTRAINT DEPTMGRFK
  FOREIGN KEY (MGRSSN) REFERENCES EMPLOYEE
  ON DELETE SET DEFAULT ON UPDATE CASCADE);
```

- The name of a schema can be explicitly specified in the definition of a table:

```
CREATE TABLE   COMPANY.DEPARTMENT …
```

# 2.5. DROP SCHEMA Command

- A schema can be deleted using the **DROP SCHEMA** command. Two options: **CASCADE** and **RESTRICT**.

  If RESTRICT is specified, the operation will fail if the schema is nonempty; otherwise it will succeed and the schema will be destroyed.

  If CASCADE is specified, the operation will always succeed.

  **DROP SCHEMA** COMPANY **CASCADE;**

# 2.6. DROP TABLE Command

- A table can be deleted using the **DROP TABLE** command. Two options: **CASCADE** and **RESTRICT**.

  If RESTRICT is specified, the operation will succeed only if the table is not referenced in any constraints or views.

  If CASCADE is specified, the operation will always succeed and propagate to all constraints and views that reference the table.

**DROP TABLE** DEPENDENT **CASCADE;**

# 2.7. ALTER TABLE Command (1)

- Adding a column.

**ALTER TABLE** COMPANY.EMPLOYEE **ADD JOB VARCHAR(12);**

**ALTER TABLE** COMPANY.EMPLOYEE **ADD JOB VARCHAR(12) NOT NULL;**

A DEFAULT clause can also be specified. If no such clause is specified, the new attribute will have NULLs in all the tuples of the relation after the command is executed.

- Dropping a column.

**ALTER TABLE** COMPANY.EMPLOYEE **DROP** ADDRESS **CASCADE;**

CASCADE option: all constraints and views that reference the column are dropped from the schema.
RESTRICT option: the command is successful only if no views or constraints reference the column.

# 2.7. ALTER TABLE Command (2)

- Dropping a default clause in a column.

  **ALTER TABLE** COMPANY.DEPARTMENT **ALTER** MGRSSN **DROP DEFAULT;**

- Adding a default clause in a column.

  **ALTER TABLE** COMPANY.DEPARTMENT **ALTER** MGRSSN **SET DEFAULT '33344555';**

# 2.7. ALTER TABLE Command (3)

- Dropping a constraint.

  To be dropped a constraint must have a name.

  **ALTER TABLE** COMPANY.EMPLOYEE **DROP CONSTRAINT** EMPSUPERFK **CASCADE;**

  *CASCADE and RESTRICT have meaning only if the constraint is a candidate key definition*. In this case, if CASCADE is specified, all the foreign key definitions that reference the candidate key will be dropped too.

- Adding a constraint.

  **ALTER TABLE** COMPANY.EMPLOYEE **ADD CONSTRAINT** EMPSUPERFK **FOREIGN KEY** SUPERSSN **REFERENCES** EMPLOYEE**;**

  The constraint to be created can be named or unnamed.

# 2.8. CIRCULAR FOREIGN KEYS (1)
# the chicken-and-egg problem

- Defining foreign keys in tables raises some other issues:
- Consider the table definitions:

```
CREATE TABLE   EMPLOYEE
…
FOREIGN KEY (DNO) REFERENCES  DEPARTMENT(DNUMBER)
```

and

```
CREATE TABLE   DEPARTMENT
…
FOREIGN KEY (MGRSSN) REFERENCES EMPLOYEE(SSN)
```

- The problem is that either EMPLOYEE or DEPARTMENT table has to be defined first.
  In both cases the system will issue an error message.
  Why?

# 2.8. CIRCULAR FOREIGN KEYS (2)
# the chicken-and-egg problem

- One solution is to postpone the introduction of the foreign-key constraint in the first table.

- For instance, if CREATE TABLE EMPLOYEE is executed first, we should not have the foreign key clause FOREIGN KEY (DNO) … in the statement.

- After CREATE TABLE DEPARTMENT has been processed, we can add the desired constraint to employee using the ALTER TABLE command:

```
ALTER TABLE EMPLOYEE
ADD CONSTRAINT EMPDEPTFK
FOREIGN KEY (DNO) REFERENCES DEPARTMENT
…
```

# 2.8. CIRCULAR FOREIGN KEYS (3)
# the chicken-and-egg problem

- If we want to populate our COMPANY database, we are in front of another problem.

- If we try to insert a tuple to relation EMPLOYEE with any non-NULL value for attribute DNO, *our insertion will be rejected as it violates the foreign key constraint*
  DNO references DEPARTMENT(DNUMBER).

  If we try to insert a tuple to relation DEPARTMENT with any non-NULL value for attribute MGRSSN, *our insertion will be rejected as it violates the foreign key constraint*
  MGRSSN references EMPLOYEE(SSN).

- We are again facing the chicken-and-egg problem!

# 2.8. CIRCULAR FOREIGN KEYS (4)
# the chicken-and-egg problem

- One solution is to initially replace DNO component in all tuples to be inserted in the EMPLOYEE relation with NULL. Then, when DEPARTMENT is populated with appropriate tuples, we can scan relation EMPLOYEE and replace NULLs with valid department numbers.

- This solution is *awkward* and *error-prone*.

- A better solution is to use *transactions* and *deferred checking of integrity constraints*. We will discuss this issue later.